

Assessment Task 2

ICT30120_BSBXCS303_AT2_TQM_v4.docx



Student Name		Student Number	
Unit Code/s & Name/s	BSBXCS303 Securely manage personally identifiable information and workplace information		
Cluster Name <i>If applicable</i>	N/A		
Assessment Name	Portfolio	Assessment Task No.	2 of 2
Assessment Due Date	Week 8		
Assessor Name	Lyndon Vella		
Student Declaration: I declare that this assessment is my own work. Any ideas and comments made by other people have been acknowledged as references. I understand that if this statement is found to be false, it will be regarded as misconduct and will be subject to disciplinary action as outlined in the TAFE Queensland Student Rules. I understand that by emailing or submitting this assessment electronically, I agree to this Declaration in lieu of a written signature.			
Student Signature		Date	

Instructions to Student	General Instructions: You are required to complete Parts 1 – 8 below (starting Page 3 of this document). There will be a variety of practical tasks that require screenshots, explanations, completing templates, checklists and filling out tables. This is an individual assessment. All answers must be in your own words and screenshots must have your name and student number shown within the screenshot. Example: Have a text document open with your name and student number to verify it is your work. Materials to be supplied by Student: Access to a computer with: - Internet access - Web browser - Microsoft office Work, Health and Safety: A work health and safety check of the assessment environment is to be conducted prior to the assessment and any hazards addressed appropriately. Assessment Criteria: To achieve a satisfactory result, your assessor will be looking for your ability to demonstrate the following key skills/tasks/knowledge to an acceptable industry standard: - Classify and label sensitive data according to Anytown Marketing policies - Encrypt device hard drives - Implement access controls to sensitive information - Confirm data integrity by comparing hash values before and after transfer - Backup sensitive data on-site and off-site - Complete a Privacy Impact Statement - Complete a Data Protection Compliance Checklist for GDPR Number of Attempts: You will receive up to two (2) attempts at this assessment task. Should your 1st attempt be unsatisfactory (U), your teacher will provide feedback and discuss the relevant sections / questions with you and will arrange a due date for the submission of your 2nd attempt. If your 2nd submission is unsatisfactory (U), or
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	you fail to submit a 2nd attempt, you will receive an overall unsatisfactory result for this assessment task. Only one re-assessment attempt may be granted for each assessment task. <i>For more information, refer to the Student Rules.</i>
Submission details	Insert your details on page 1 and sign the Student Declaration. Include this form with your submission. Due Date: Week 8 Submission Details: You are required to complete the tasks in this document in the space provided under each and save the document as BSBXCS303_AT2_<Your Name> You are also required to complete the following documents and submit them with your completed Portfolio. ICT30120_BSBXCS303_LHO_PrivacyImpactStatement_Template ICT30120_BSBXCS303_LHO_DataProtectionChecklist Once completed, upload the document to the Assessment Task 1 within the BSBXCS303 Unit on Connect. TAFE Queensland Learning Management System: Connect URL: https://connect.tafeqld.edu.au/d2l/login • Username; 9-digit student number • For Password: Reset password go to: https://passwordreset.tafeqld.edu.au/default.aspx
Instructions to Assessor	Specifications of assessment: There are 8 parts including 1 template and 1 checklist as listed below, that must all be completed correctly to achieve a satisfactory result for this assessment. Equipment or material requirements: Documents: - ICT30120_BSBXCS303_LHO_AnytownMarketingPolicies - ICT30120_BSBXCS303_LHO_StaffInformation - ICT30120_BSBXCS303_LHO_PrivacyImpactAssessment_Template - ICT30120_BSBXCS303_LHO_DataProtectionChecklist - ICT30120_BSBXCS303_LHO_StoredFileDescription - Access to a computer with: - o Internet access - o Web browser - o Microsoft office - Access to an AWS Learner Lab account

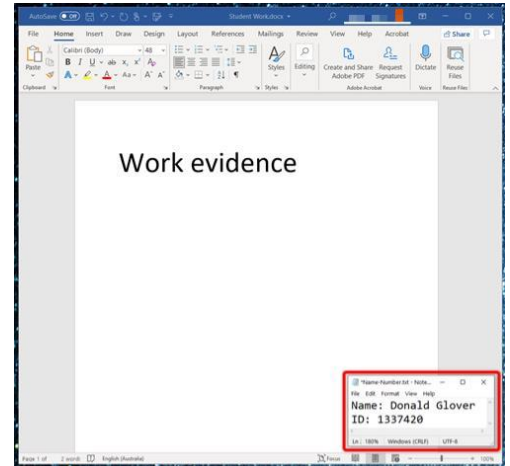
	Details of location: TAFE will provide a simulated work environment in the classroom; however, it is possible to complete these tasks at home using a computer with internet access and the correct software. Time restrictions: This is a Portfolio assessment designed to take place over 12 weeks. Level of assistance permitted: Teachers and/or Tutors should be available in class, and accessible by email for students working from home. Staff cannot directly provide students answers but can guide them to complete tasks individually. Work Health and Safety: A work health and safety check of the assessment environment is to be conducted prior to the assessment and any hazards addressed appropriately. Interactions with team members: This is an individual assessment and all assessment submissions should be completed individually. Though some consultation is allowed between the cohort. Contingencies if conditions cannot be met: Students will be able to resubmit their work a second time after feedback has been provided if they are not successful on the first attempt. Reasonable adjustments will also be made for students as and when appropriate, after consultation with the Accessibility and Counselling team. – You must see your teacher prior to assessment regarding this.
Note to Student	An overview of all Assessment Tasks relevant to this unit is located in the Unit Study Guide.

Scenario: You are employed by Anytown Marketing as an IT consultant to assist in implementing their data security and privacy handling policies and procedures.

See “**ICT30120_BSBXCS303_LHO_AnytownMarketingPolicies**”. This implementation project is to be completed with the required documentation submitted to the authorising manager.

Note, it's required to set up a Virtual Server on the AWS cloud, using AWS Learner labs. You will be provided with an invitation to join AWS Academy Learner Labs which you must accept to get access the AWS cloud. No credit card information is required for Learner Labs.

All visual evidence (screenshots) must include your student details such as Student Name and Student Number (ID). To achieve this, you can use a screenshot tool (for Windows, WIN + SHIFT + S) and overlay text, or simply have a text file open as you take the screenshot.



For example:

Part 1

You are required to classify **ALL (12) the Anytown Marketing Firms' files** into their data classification types and new storage locations. Use the “**ICT30120_BSBXCS303_LHO_StoredFileDescription**” documents for the file names, types, and description of data within.

Refer to the Anytown Marketing Policy, page 1 -2 to review how documents and information should be classified.

Note, Label Colours must be included in the table below, but do not need to be implemented on files and folders.

Review the information in the file descriptions in StoredFileDescription and consider the following questions along with the classification rules outlined in the AnyTown Marketing Policy.

- *Does the file contain PII Sensitive information?*
- *Does the file contain PII Confidential information?*
- *Does the file contain business secrets and proprietary information?*
- *Does the file contain business information?*
- *Does the file contain information that should be available to the public?*

If you are unsure, make your best decision and explain the reason for your decision.

Classification:	Location	Label Colour
Public	(Website/File Server/	(None/Green/Yellow/Red)

	Secure File Server)	
<i>Name of file</i>	<i>Add rows as necessary</i>	
Classification: Internal	Location (Website/File Server/ Secure File Server)	Label Colour (None/Green/Yellow/Red)
<i>Name of file</i>	<i>Add rows as necessary</i>	
Classification: Confidential	Location (Website/File Server/ Secure File Server)	Label Colour (None/Green/Yellow/Red)
<i>Name of file</i>	<i>Add rows as necessary</i>	
Classification: Confidential/Sensitive	Location (Website/File Server/ Secure File Server)	Label Colour (None/Green/Yellow/Red)
<i>Name of file</i>	<i>Add rows as necessary</i>	

Part 2

You are required to provide a short explanation (~50 words) on how encryption protects sensitive data.

You then must also demonstrate encrypting data and provide screenshots as evidence. You can do this in two methods:

- 1) add a [Virtual Hard Drive](#) to your Virtual Machine and encrypt it using [Bitlocker](#) or,
- 2) encrypt the Confidential/Sensitive folder with 2 test files using a third-party encryption tool such as [7Zip](#).

WARNING! Do not encrypt your personal computer's Hard Drive, you must use a Virtual Machine.

Explanation	<i>At least 50 words are expected. Explain how encryption works using words like plain text, cipher text, key and algorithm. Provide an example of encryption that you have used. Explain how to make encryption less susceptible to brute force attack.</i>
Screenshot(s)	• <i>Screenshot of 2nd hard disk locked by BitLocker</i> • <i>Or Screenshot of the Confidential/Sensitive folder with 2 files inside it, zipped with 7 zip.</i>

Part 3

You are required to give a brief explanation (~50 words) for the best practices for sharing sensitive data within the workplace.

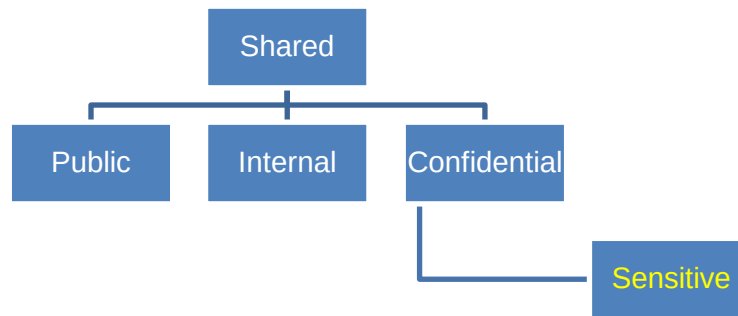
You then must demonstrate the following and provide screenshots as evidence of completion:

Requirements:

- 1) Setup a User Group called MarketingManagers
- 2) Create 5 user accounts as follows:

First Name	Last Name	Username	Password
Felix	Randall	randf	P@ssw0rd
Cornell	Stacks	stacc	
Nona	Benton	bentn	
Jayson	Nielson	jaysn	
Fabian	Rogerson	fabir	

- 3) Add the first two accounts (Felix Randall and Stacks Cornell) to the MarketingManager group.
- 4) Create the folder structure described in AnyTown Marketing Policies. Create 1 or more files for each folder following your classification in Q1.



- 5) Configure the Shared folder as a network share and provide Staff, read access to the Shared folder over the network.
- 6) Configure Security permissions on the Confidential folder and provide modify access to the MarketingManagers group.
- 7) Once the folder has been shared, encrypt the folder and all files within.

Explanation	<i>At least 50 words are expected.</i> <i>Provide 5 best practices for sharing sensitive information that include technical controls and/or people procedures.</i> <i>Do some research on best practice for sharing sensitive information and consider the steps you have taken in the hands-on handouts.</i>
Screenshot(s)	• <i>Screenshot of 5 user accounts created.</i> • <i>Screenshot of marketing_department group showing the five user accounts added.</i> • <i>Screenshot of folders created (either in file explorer or in command with tree command)</i> • <i>Screenshot of security permissions provided to the marketing_department group on the confidential folder.</i> • <i>Screenshot to confirm the share has been created - use net share or use shares in shared folders in computer management.</i> • <i>Screenshot of the sensitive folder using EFS or 7zip,</i> <i>Opportunity for improvement:</i> - <i>Test access to the share</i> - <i>Test access to the confidential folder</i> - <i>Test access to the sensitive folder</i>

Part 4

You must perform a hash calculation on the **"ICT30120_BSBXCS303_LHO_StaffInformation"** document using a hashing algorithm, for example SHA512. Then transfer the file into the encrypted shared folder you created in **Part 3** and run the hash calculation again to ensure data integrity after the transfer.

In your Windows 10 Virtual Machine, use one of the two following methods:

- 1) a third-party tool such as [7Zip](#) and [HashCalc](#) or,
- 2) a terminal emulator such as [PowerShell](#).

Ensure to include visual evidence of the files hash before the transfer, and after the transfer is complete, multiple screenshots will be required.

Screenshot(s)	• <i>Copy the StaffInformation.pdf file to the downloads folder or create a new file and copy the contents of StaffInformation.pdf into the file.</i> • <i>Check the Anytown policy for the hash required.</i> • <i>Take a screenshot of the hash of the file in the downloads folder.</i> • <i>Move the file to the Sensitive folder.</i> • <i>Take a screenshot of the hash of the file in the sensitive folder.</i> • <i>Hashes should match.</i> <i>Add screenshot of whole hash in the original folder (downloads folder)</i> <i>Add screenshot of whole hash in the destination folder (sensitive folder)</i>
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Part 5

You are required to explain an attack that would compromise the integrity of sensitive data.

You then must demonstrate setting up file change audits via your Windows 10 Virtual Machine. This is to keep track of changes made to the sensitive data file used in **Part 4** to report if attacks have occurred. Provide screenshots as evidence of completion:

Explanation	<i>Key word here is integrity. Write a sentence on what integrity means and why you don't want it to be compromised.</i> <i>Research an attack that would compromise file integrity. Explain how the attack changes the integrity of the file.</i> <i>Explain how you would be able to detect that integrity has changed. Use an example to explain. Screenshots can help you explain too.</i> <i>Tell the story of the attack, what the attack did to the file and how you would know what the attack did.</i>
Screenshot(s)	Screenshots required: • <i>Screenshot of the local security policy for object access showing success and failure ticked.</i> • <i>Screenshot of properties, advanced, auditing on the folder that is getting audited.</i> • <i>Screenshot of log in event viewer showing the user accessing the folder (4663 ID or 5145)</i>

Part 6

You are required to explain how you can keep the data integrity intact and secure during transport and at the off-site location.

You then must demonstrate and provide screenshots for backing up the encrypted folder created in **Part 3** according to organisational procedures

"**ICT30120_BSBXCS303_LHO_AnytownMarketingPolicies**". This is required to be backed up to removable storage for transport and storage off-site.

In your Windows 10 Virtual Machine, add a [Virtual Hard Drive](#), then use one of the two following methods to backup data:

- 1) Backup and Restore (Windows 7) or,
- 2) a third-party tool such as [FreeFileSync](#).

Explanation	- <i>At least 50 words</i> - <i>Looking for 5 best practices for protecting integrity for data in motion – over the network and physical transport and data at rest at the off-site location.</i> - <i>Can use dot points but explain each point.</i> - <i>Consider technical controls and procedures (people controls)</i> - <i>Looking for a defence in depth answer</i>
Screenshot(s)	- <i>A configuration setting of the backup/sync job using windows backup of FreeFileSync</i> - <i>A snip showing what was copied in Windows Backup or FreeFileSync</i> - o <i>Look for a log of the backup/sync</i>

Part 7

You are required to complete a Privacy Impact Assessment

“ICT30120_BSBXCS303_LHO_PrivacyImpactAssessment_Template”. See the PIA template to be completed and submitted with this document.

- *See the PIA guide.pdf*

Part 8

You are required to complete the Data Protection & Compliance Checklist to ensure GDPR compliance **“ICT30120_BSBXCS303_LHO_DataProtectionChecklist”**. The checklist is to be submitted with this document.

- *Assume the recommendations you made in the PIA have been implemented.*
- *Read each criteria in the Data Protection and Compliance checklist and tick the box if you think the criteria has been met.*

***** END OF ASSESSMENT TASK *****